

Running Enrollment Count Per Student By Course

✓ Database Description

The **University Core** database is designed to manage and organize essential academic information within a university setting. It maintains comprehensive records of departments, students, courses, and enrollments. The system facilitates the tracking of student-course registrations, departmental offerings, and the grading history of students, ensuring efficient academic administration and reporting

✓ Schema Specification

📋 Departments

Stores all academic departments within the university.

Column Name	Data Type	Constraints	Description
department_id	INT	PRIMARY KEY	Unique identifier for department
department_name	VARCHAR(50)	NOT NULL	Name of the department

📋 Students

Stores student information along with their department.

Column Name	Data Type	Constraints	Description
student_id	INT	PRIMARY KEY	Unique student ID
name	VARCHAR(100)	NOT NULL	Full name of the student
age	INT		Age of the student
department_id	INT	FOREIGN KEY → Departments(department_id)	Student's home department

📋 Courses

Stores the list of courses offered by departments.

Column Name	Data Type	Constraints	Description
course_id	VARCHAR(10)	PRIMARY KEY	Unique course identifier
course_name	VARCHAR(100)	NOT NULL	Name of the course
department_id	INT	FOREIGN KEY → Departments(department_id)	Department offering the course

Enrollments

Stores student enrollments in courses along with their grades.

Column Name	Data Type	Constraints	Description
student_id	INT	FOREIGN KEY → Students(student_id)	Enrolled student ID
course_id	VARCHAR(10)	FOREIGN KEY → Courses(course_id)	Course enrolled in
grade	VARCHAR(5)		Grade achieved in the course
PRIMARY KEY	Composite	PRIMARY KEY (student_id, course_id)	Composite key to prevent duplicates

✓ Sample Data

📖 Departments

department_id	department_name
1	Computer Science
2	Mathematics
3	Physics
4	Literature

📖 Students

student_id	name	age	department_id
101	Alice Smith	20	1
102	Bob Johnson	22	2
103	Clara Evans	21	3
104	David Kumar	23	1

📖 Courses

course_id	course_name	department_id
C101	Algorithms	1
C102	Linear Algebra	2
C103	Quantum Mechanics	3
C104	Shakespearean Drama	4

Enrollments

student_id	course_id	grade
101	C101	A
101	C102	B+
101	C104	A-
102	C102	B
102	C103	A
103	C103	B-
103	C104	A
104	C101	A+
104	C103	B
104	C102	C+
104	C104	B
101	C103	B+

Problem Statement

Write a query to calculate a **running total of enrollments** for each student, based on the **alphabetical order of the course_id**. For each enrollment, show:

- student_id
- name
- course_id
- running_enrollment_count — number of courses enrolled by that student up to and including that course (based on course_id order)

Use a window function with PARTITION BY student_id and ORDER BY course_id.

Expected Output

student_id	name	course_id	running_enrollment_count
101	Alice Smith	C101	1
101	Alice Smith	C102	2
101	Alice Smith	C103	3
101	Alice Smith	C104	4
102	Bob Johnson	C102	1
102	Bob Johnson	C103	2
103	Clara Evans	C103	1
103	Clara Evans	C104	2
104	David Kumar	C101	1
104	David Kumar	C102	2
104	David Kumar	C103	3
104	David Kumar	C104	4